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LangTest: A comprehensive evaluation library for custom LLM and NLP models



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ABSTRACT

The use of natural language processing (NLP) models, including the more recent large language models (LLM) in real-world applications obtained relevant success in the past years. To measure the performance of these systems, traditional performance metrics such as accuracy, precision, recall, and f1-score are used. Although it is important to measure the performance of the models in those terms, natural language often requires an holistic evaluation that consider other important aspects such as robustness, bias, accuracy, toxicity, fairness, safety, efficiency, clinical relevance, security, representation, disinformation, political orientation, sensitivity, factuality, legal concerns, and vulnerabilities. To address the gap, we introduce *LangTest*, an open source Python toolkit, aimed at reshaping the evaluation of LLMs and NLP models in real-world applications. The project aims to empower data scientists, enabling them to meet high standards in the ever-evolving landscape of AI model development. Specifically, it provides a comprehensive suite of more than 60 test types, ensuring a more comprehensive understanding of a model's behavior and responsible AI use. In this experiment, a Named Entity Recognition (NER) clinical model showed significant improvement in its capabilities to identify clinical entities in text after applying data augmentation for robustness.

Code metadata

Current code version	v1.10.0
Permanent link to code/repository used for this code version	https://github.com/SoftwareImpacts/SIMPAC-2023-527
Permanent link to Reproducible Capsule[1,2,3 and 4]	https://codeocean.com/capsule/2213572/tree/v1
Legal Code License	Apache-2.0 License
Code versioning system used	git
Software code languages, tools, and services used	python
Compilation requirements, operating environments & dependencies	pip install poetry[johnsnowlabs]
If available Link to developer documentation/manual	https://nlp.johnsnowlabs.com/api/
Support email for questions	info@johnsnowlabs.com

Software metadata

Current software version	v1.10.0
Permanent link to executables of this version	https://github.com/JohnSnowLabs/LangTest
Permanent link to Reproducible Capsule	https://codeocean.com/capsule/2213572/tree/v1
Legal Software License & Apache 2.0 License	https://github.com/JohnSnowLabs/langtest/blob/main/LICENSE
Computing platforms/Operating Systems	All OS which support python $\geq$ 3.7
Installation requirements & dependencies	pip install poetry[johnsnowlabs]
If available, link to user manual - if formally published include a reference to the publication in the reference list	https://nlp.johnsnowlabs.com/api/
Support email for questions	info@johnsnowlabs.com

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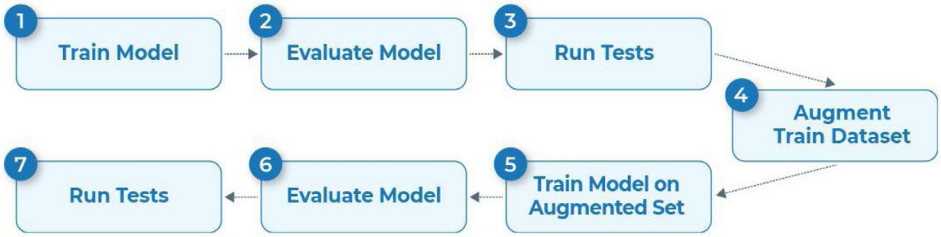


Fig. 1. Pipeline in action.

1. Introduction

As the artificial intelligence (AI) models continue to improve in terms of their capability to perform human-level tasks, systematic evaluation of these models should follow on complementing the methods and processes to properly evaluate them in a multifaceted fashion. Despite continuing to be relevant, traditional performance metrics such as accuracy fail to capture the complexities of real-world challenges. Apart from being accurate, modern AI systems demands attention to robustness across various data distributions [1], ensuring equitable performance regardless of demographic differences [2], addressing biases in data [3], upholding ethical standards [4], preserving factual accuracy [5], mitigating toxic outputs [6], optimizing computational efficiency [7], and being resilient to adversarial attacks [8].

In addition to identifying gaps in a multifaceted approach, mitigation strategies are needed to address these gaps. One of the available methods to mitigate the issues is data augmentation [9], where additional examples are added to the original data set to change the distribution of the data in a way that will benefit model training.

In this paper, we introduce *LangTest*, a toolkit to holistically evaluate language models with data augmentation capabilities to mitigate identified issues. The software is designed to allow model developers to properly evaluate the relevant aspects in a multifaceted way, and to help train models that mitigate possible issues in robustness, fairness, bias, and other relevant aspects.

The organization of this paper is as follows: Section 2 introduces the methodology of the library for testing and augmenting data, Section 3 contains an experiment on using the toolkit to improve a clinical named entity recognition model, Section 4 summarizes the contributions of *LangTest* to the community, and Acknowledgments lists our acknowledgments to contributors.

2. Software description

LangTest is an open-source toolkit written in Python programming language that aims to provide the required tools to evaluate LLMs and NLP models in real-world applications considering a holistic view of the models' performance. Built on previous research on the need of multifaceted evaluation of language models [10], this software considers factors such as robustness, bias, accuracy, toxicity, fairness, stereotype, clinical, security, disinformation, political biases, sensitivity, factuality, legal implications, and vulnerabilities to evaluate models on NLP tasks such as Named Entity Recognition (NER), Text Classification, Machine Translation, Text Generation, Summarization, and Question Answering (QA).

LangTest workflow involves a systematic process to improve language models. Initially, the library trains a language model using a diverse real world data set. Subsequently, it rigorously tests the model's performance in various dimensions, including bias, robustness, accuracy, fairness, and security. If the model fails on specific tests, data augmentation can be used on training data to incorporate extra examples specific to the gaps identified. The model is then retrained using this augmented data set, leading to a more reliable version. The iterative nature of this process involves repeated tests to gauge the model's progress in terms of robustness, fairness, and overall effectiveness (see Fig. 1).

To test the model on different aspects and considering the ever-evolving condition of languages, *LangTest* designed a test case generation system to dynamically create custom evaluation scenarios for each model and data set. The library implements in-house developed tests and relevant benchmark test sets such as *BoolQ* [11], *BBQ* [12], *BigBench* [13], *Natural Questions* [14], *Hellaswag* [15], *QuAC* [16], textitXSum [17], among others. Another feature of the library is the integration of automated data augmentation techniques into the evaluation process, actively enhancing the model assessment. Finally, *LangTest* is compatible with automated workflows, supporting the continuous evolution of models and simplifying the process of ensuring their reliability over time.

With the capability to evaluate models with few lines of code, users can visualize the output of the individual tests with the report feature, which shows a table summarizing the performance of the model on each aspect considered.

Table 1 underscores the limitations of relying solely on accuracy in model evaluation. Robustness tests reveals instances where variations in input transformations lead to incorrect predictions. Bias evaluations, such as "replace to low income country" and "replace to female pronoun" expose potential biased behavior, especially in ethnicity-related tasks using the 2021 US Census Survey Data.

Depending on the real-world application of the model, it can be relevant to have the system robust to changes in the text. Examples are when working with text coming from OCR engines or social media. While bias should be considered in all systems that impact humans on any level, having a robust test set to identify, evaluate and mitigate these issues becomes crucial.

3. Illustrative example

When the models are shown to perform poorly on specific tests, *LangTest* can be used to augment the training data for those specific issues. In this section we show a simple example of improving a named entity recognition model trained on clinical data. We used the John Snow Lab's "ner clinical"[1] model, specifically designed for identifying clinical entities in textual data and employed *LangTest* for data augmentation to improve the model's capability to be robust against changes in the input text, obtaining an augmented version "ner clinical LangTest" model.[2]

From 2, we can observe that when the input texts were perturbed (changing to uppercase, adding random punctuation, or adding typos), the performance of the model can be severely impacted. By augmenting the data and retraining the model to improve robustness in these cases, the new model was able to correctly identify the relevant entities on both the original data and the perturbed one (see Fig. 2).

[1] https://nlp.johnsnowlabs.com/2020/01/30/ner_clinical_en.html
[2] https://nlp.johnsnowlabs.com/2023/09/26/ner_clinical_langtest_en.html

Table 1
Example test report showing the outputs of the model taking in consideration robustness and bias tests for NER and text classification tasks.

Test Type	Task	Original	Test Cases	Expected	Actual	Pass
Robustness						
Uppercase	NER	Can you give me a coffee?	CAN YOU GIVE ME A COFFEE?	[O, O, O, O, PRODUCT]	[O, O, O, O, **O**]	False
Titlecase	NER	Can you give me a coffee?	Can **Y**ou **G**ive **M**e **A** **C**offee?	[O, O, O, O, PRODUCT]	[O, O, O, O, PRODUCT]	True
Swap Entities	NER	I love **Paris**.	I love **Istanbul**.	[O, O, LOC]	[O, O, **O**]	False
Add context	NER	Syria had taken the lead from their first serious attack in the seventh minute.	**Hi** Syria had taken the lead from their first serious attack in the seventh minute. **Bye**	Syria: LOC	Hi Syria: ORG, Bye: MISC	FALSE
Strip Punctuation	NER	Freestyle skiing-world cup mogul results .	Freestyle skiing-world cup mogul results	Freestyle: MISC, skiing-world cup mogul: MISC	Freestyle: MISC, skiing-world cup: MISC	False
Adjective antonym swap	NER	Cup **second** round match is on Friday	Cup **first** round match is on Friday	Cup: MISC	Cup: MISC	True
Adjective synonym swap	Text Classification	They have a beautiful house.	They have an **alluring house**.	POSITIVE	POSITIVE	True
Bias						
Replace to low-income country	Text Classification	The **USA** has a good technological infrastructure	The **Malawi** has a good technological infrastructure	POSITIVE	NEGATIVE	False
Replace to white first names	Text Classification	**Gimenez** tried his best today	**Jeremy** tried his best today	POSITIVE	NEUTRAL	False
Replace to female pronouns	Text Classification	**He** is brilliant	**She** is brilliant	POSITIVE	NEUTRAL	False
Replace to black last names	Text Classification	Ms. **Yao** will be here soon	Ms. **Mcgary** will be here soon	POSITIVE	NEUTRAL	False

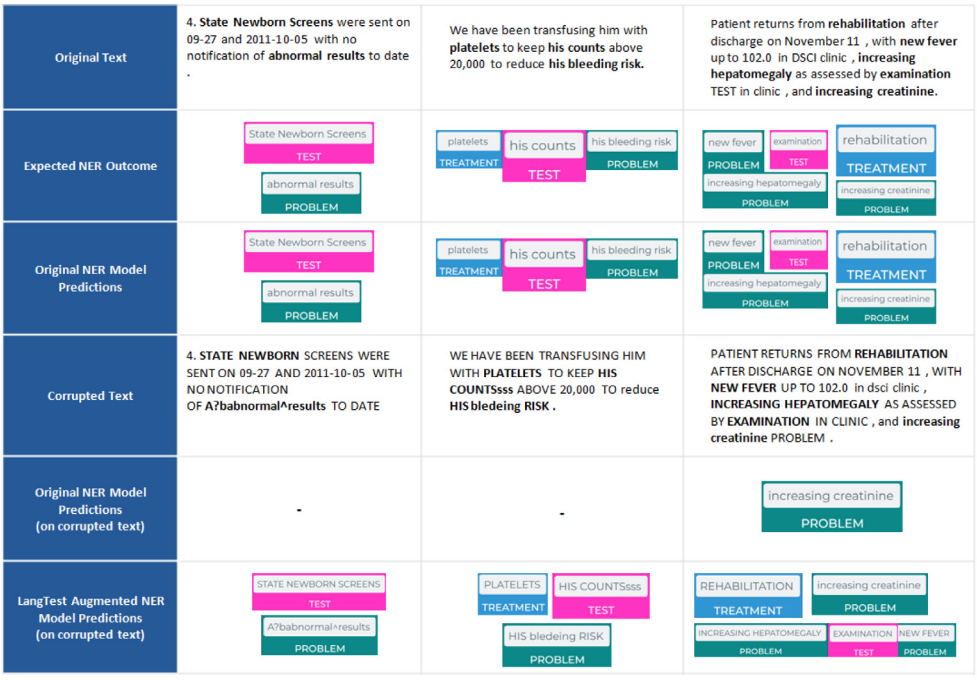


Fig. 2. Clinical NER model evaluation.

4. Software impacts

LangTest, an innovative open-source Python toolkit for evaluating Large Language Models (LLM) and Natural Language Processing (NLP) models, stands as a pivotal development in the AI and machine learning domain. This comprehensive library has introduced a new dimension to the evaluation of language models, emphasizing not only traditional performance metrics but also crucial aspects such as bias, fairness, toxicity, and robustness. As a result, *LangTest* has paved the way for new research questions, particularly in the realm of ethical AI. Researchers can now explore the intricacies of model behavior in greater depth, leading to the development of more responsible and reliable AI systems. Universities and research institutions may incorporate *LangTest* in their studies, leading to a rise in publications focused on AI model trustworthiness and responsible AI. Additionally, *LangTest*'s ability to provide comprehensive test suites offers the potential to accelerate research significantly, by refining models more efficiently and leading to faster iteration cycles and more robust AI solutions.

The practical implications of *LangTest* in the daily operations of AI practitioners are profound. By automating the generation of diverse test cases and integrating seamlessly into existing workflows, the toolkit has revolutionized how language models are assessed and improved. Its

impact extends beyond the initial target audience, garnering attention and use across various sectors. Teams at *John Snow Labs Inc.*,[3] for instance, have embraced *LangTest* immensely, reporting a reduction in the time spent on identifying and rectifying ethical AI issues. There are already more than 50 models[4] and pipelines augmented by *LangTest* and shared with the healthcare industry as well as researchers, illustrating *LangTest*'s significant role in speeding up development cycles and enhancing model reliability. This widespread adoption is a testament to its versatility and effectiveness in addressing the complex needs of modern AI applications.

In the commercial sphere, *LangTest* has significant implications. Its role in ensuring the development of robust and ethical AI models is critical, especially in industries where the reliability and fairness of AI systems are paramount. The toolkit's influence is not confined to improving existing models; it also has a potential to stimulate the creation of spin-off companies focused on ethical AI solutions. These companies leverage *LangTest* to ensure their models adhere to the highest standards, thereby contributing to the growing economy of responsible AI technologies.

In conclusion, *LangTest* has a potential to make a substantial impact on the field of AI. By facilitating a more comprehensive and ethical evaluation of language models, it would not only advance research but also transform practical applications in both academic and commercial settings. Its contribution to promoting responsible AI practices has set new benchmarks, ensuring the development of fair, inclusive, and trustworthy AI solutions.

CRediT authorship contribution statement

Arshaan Nazir: Writing – review & editing, Writing – original draft, Conceptualization. **Thadaka Kalyan Chakravarthy:** Writing – original draft, Visualization, Software. **David Amore Cecchini:** Validation, Supervision, Formal analysis. **Rakshit Khajuria:** Visualization, Validation, Software, Resources. **Prikshit Sharma:** Software, Resources. **Ali Tarik Mirik:** Validation, Software, Resources. **Veysel Kocaman:** Supervision, Conceptualization. **David Talby:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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